Supplementary Material: Unlimited Number of Spanning Clusters in 3D Percolation Above the Critical Point

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I. RULES OF THE MODEL

Figure 1 illustrates the rules of our model.

II. SECOND MOMENT

Figure 2 shows that the curves for the second moment χ and the size of the largest cluster S present some dependence on the percolation order i.

Figure 3 and 4 illustrate the evolution of cluster properties for a simple cubic lattice and a cubic lattice with bonds between nearest neighbors and second nearest neighbors (2NN), respectively, in the case of eight species. The inset shows that the curves of the second moment χ for different values of L collapse on top of each other when the data is rescaled using the critical exponents $\gamma = 1.819 \pm 0.003$ [1] and $\nu = 0.875 \pm 0.001$ [2] of standard 3D percolation.

III. CRITICAL POINT AS A FUNCTION OF k AND ρ

Figure 5 presents the average percolation thresholds $p_c^{(1)}$ at which the first species exhibits for the first time a cluster that reaches both the top and the bottom faces of the system for a cubic lattice with $L = 512$. We observe that the average percolation threshold grows linearly with ρ and Fig. 6 shows that the slopes of the curves grow exponentially with k.

IV. FRACTAL DIMENSION

Figure 7 shows that, at the critical point p_c, the least-squares fit to the cluster mass M_i follows a power law [3]

$$M_i = b_i L^{d_f}, \tag{1}$$

where, within error bars, the fractal dimension d_f is consistent with the fractal dimension of the spanning cluster of classical percolation in 3D, $d_f = 2.5226 \pm 0.0001$ [4]. Additionally, the prefactors b_i, decrease with the percolation order i following an exponential behavior,

$$b_i = Ae^{-Bi}, \tag{2}$$

where the least-squares fit to the data using Eq. (2) gives $A = 0.75 \pm 0.01$ and $B = 0.24 \pm 0.02$, as shown in the inset of Fig. 7 decrease with their percolation order i.

V. THE INFLUENCE OF INITIAL FRACTION ON THE NUMBER OF PERCOLATING CLUSTERS

Figure 8 presents the number of percolating clusters of different species n_c as function of the initial fraction ρ for different values for L. This figure shows that, for all values of L, it is possible to have 4 percolating clusters of different species. However, it is necessary to choose ρ below a certain value ρ_{max}, otherwise there will be less percolating clusters. When analyzing the Fig. 8, we observe a transition at ρ_{max}, where the system moves from a phase where the number of percolating clusters is k_{max} to a phase where it is zero. Additionally, we observe the presence of a finite scaling law in the vicinity of ρ_{max}, which we explore by adjusting the x-axis as shown in Fig. 8, using $\nu = 0.875 \pm 0.001$ [2] and $\rho_{max} = 0.159 \pm 0.001$.

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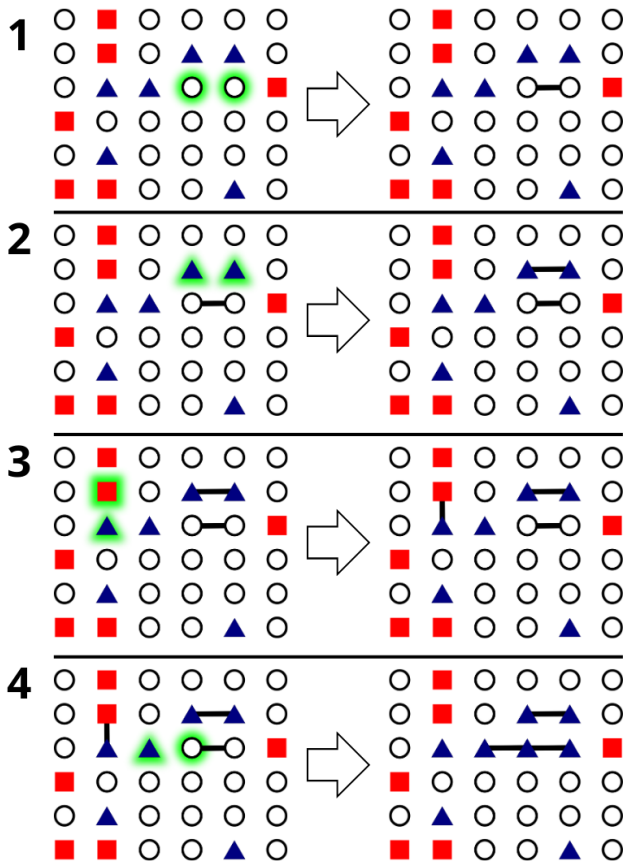

FIG. 1. This figure illustrates the rules of our model. The red squares and blue triangles represent sites inhabited by two different species, while the white circles denote uninhabited sites. On the left, the system's configuration is shown before the bond connecting the green-highlighted sites is occupied, and on the right, the system is displayed after the bond has been occupied. In Case 1, a bond connects two uninhabited sites. In Case 2, a bond connects two sites inhabited by the same species. In Case 3, a bond connects sites inhabited by different species, but both retain their species identities. In Case 4, when a bond connects an uninhabited node to a node inhabited by a certain species, all sites in the cluster adopt the species of the inhabited node.

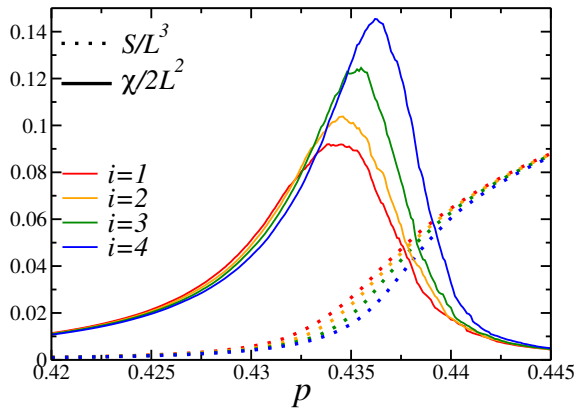

FIG. 2. Evolution of properties of the cluster distribution during the occupation process. Species are labeled in the order of percolation, that is, we label by 1 the species that first percolates, by 2 the second species to percolate and so on. The curves are the average over 1000 samples. The colors indicate the order i in which the species percolated. The dotted lines correspond to the average size S of the largest cluster, the solid lines indicate the second moment. For convenience, each of the quantities is scaled by the system size, as indicated in the figure's legend. We used a system size $L = 512$, with 4 species, and initial density $\rho = 0.025$.

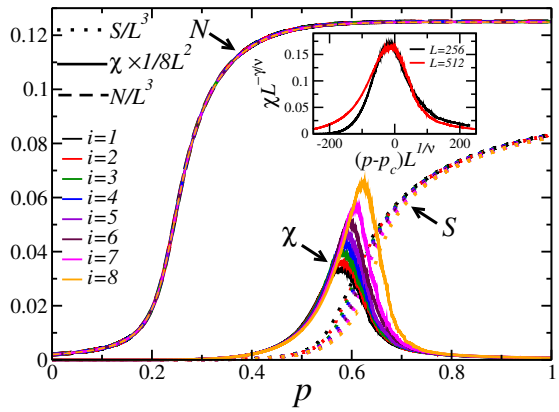

FIG. 3. Evolution of properties of the cluster distribution during the occupation process for 8 species. Species are labeled in the order of percolation, that is, we label by 1 the species that first percolates, by 2 the second species to percolate and so on. The curves are the average over 1000 samples. The colors indicate the order i in which the species percolated. The dotted lines correspond to the average size S of the largest cluster, the solid lines indicate the second moment of the cluster size distribution χ, as defined in Eq. (1) in the main text, and the dashed lines indicate the number of nodes N that have been occupied by each species. For convenience, each of the quantities is scaled by the system size, as indicated in the figure's legend. We used a system size $L = 512$, with 8 species, and initial density $\rho = 0.002$. The inset shows the finite size scaling of χ for the first percolating cluster using $p_c = 0.59$ and the critical exponents for 3D percolation.

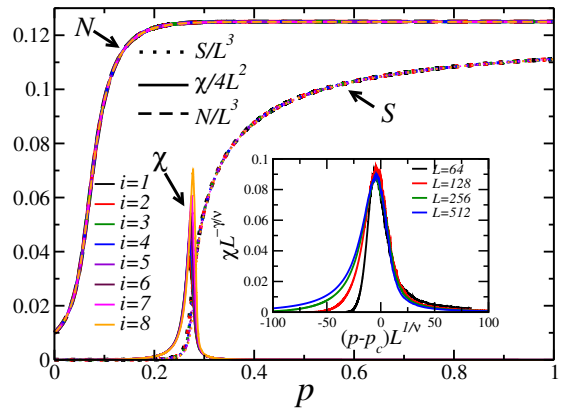

FIG. 4. Evolution of properties of the cluster distribution during the occupation process for a cubic lattice with bonds between nearest neighbors and second nearest neighbors (2NN). Species are labeled in the order of percolation, that is, we label by 1 the species that first percolates, by 2 the second species to percolate and so on. The curves are the average over 1000 samples. The colors indicate the order i in which the species percolated. The dotted lines correspond to the average size S of the largest cluster, the solid lines indicate the second moment of the cluster size distribution χ, as defined in Eq. (1) in the main text, and the dashed lines indicate the number of nodes N that have been occupied by each species. For convenience, each of the quantities is scaled by the system size, as indicated in the figure's legend. We used a system size $L = 512$, with 8 species, and initial density $\rho = 0.01$. The inset shows the finite size scaling of χ for the first percolating cluster using $p_c = 0.273$ and the critical exponents for 3D percolation.

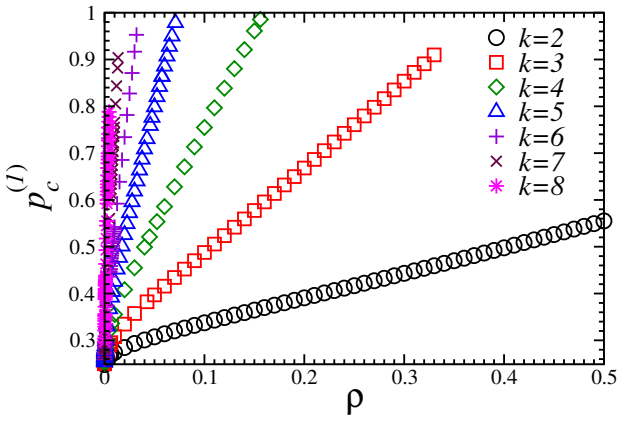

FIG. 5. Critical point of the first $p_c^{(1)}$ percolating cluster as a function of the initial fraction ρ. Each color represents a different number of k percolating clusters of different species. The average percolation threshold grows linearly with ρ and exponentially with k. The averages were performed over 100 samples of size $L = 512$.

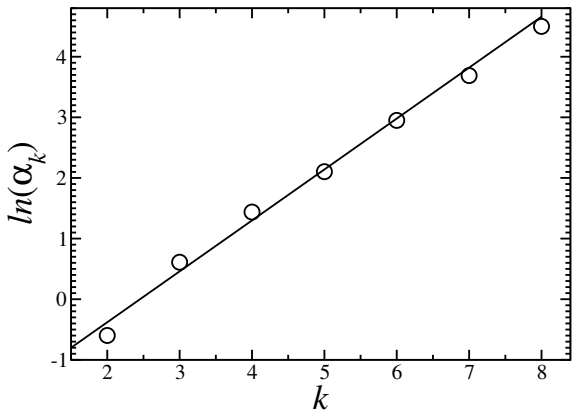

FIG. 6. Semi-logarithmic plot of the growth rate α_k of $p_c^{(1,k)}$, given that $p_c^{(1,k)} \propto \alpha_k \rho$. The rate α_k grows exponentially with k.

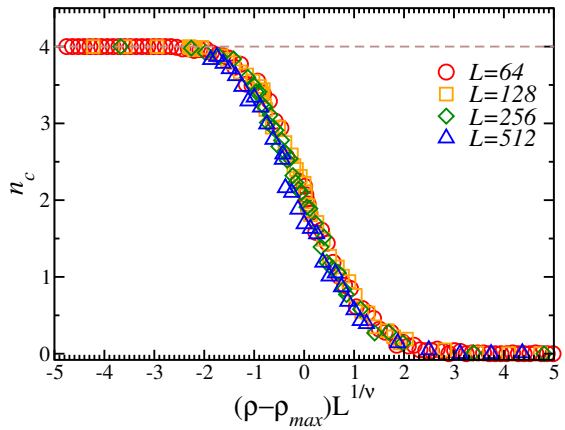

FIG. 8. Rescaled plot of average number of percolating clusters as a function of the initial fraction of inhabited nodes ρ for four species in the region around ρ_{max}. The legend indicates the side length L of each lattice. The axes show how they have been rescaled. We use a scaling exponent $\nu = 0.875 \pm 0.001$ [2] and $\rho_{max} = 0.159 \pm 0.001$.

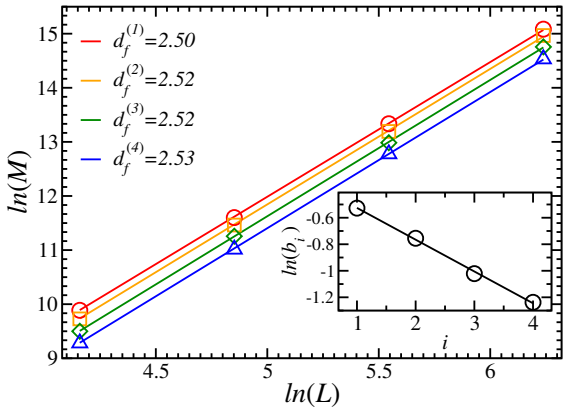

FIG. 7. Logarithmic plot of the mass M of the first (black circles), second (red squares), third (green diamonds), and fourth (blue triangles) percolating clusters at p_c as a function L. We consider four species and $\rho = 0.025$ for each species. We averaged over 1000 realizations for $L = 64$, 128, 256 and 512. The errors were found to be smaller than the size of the symbols. The fractal dimensions are obtained through the least-squares fit of the largest cluster of each species segregated by order at p_c to a power law $M = b_i L^{d_f}$, yielding the exponent $d_f^{(1)} = 2.50 \pm 0.03$, $d_f^{(2)} = 2.52 \pm 0.01$, $d_f^{(3)} = 2.52 \pm 0.01$, $d_f^{(4)} = 2.53 \pm 0.01$ for the first, second, third, and fourth percolating cluster, respectively. Within precision, our results are consistent with the fractal dimension of 3D standard percolation. In the inset we observe in a semi-logarithmic plot of the prefactors b_i decrease exponentially with i as expressed in Eq. (2).

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